

Aryan Lodha

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EDUCATION

K.J Somaiya College of Engineering

B.Tech in Artificial Intelligence & Data Science - Honours in Applied Cyber Security

Aug 2023 – Present

Mumbai, Maharashtra

EXPERIENCE

Deep Learning & Computer Vision Intern

Trueview

May 2026 – Jul 2026

Aurangabad, Maharashtra

- Developed a real-time intelligent surveillance system for restricted zone monitoring using Python, YOLOv8, ByteTrack, OpenCV, PyTorch, XGBoost, and Scikit-learn.
- Built a spatial-temporal analytics pipeline for multi-object tracking, behavioral feature extraction, and machine learning-based risk assessment to detect potential intrusions.
- Integrated real-time visualization and alert generation, achieving 89.4% behavioral classification accuracy and 87.8% F1-score, reducing false alarms by 23% versus a rule-based baseline while maintaining 28 FPS on CUDA-enabled hardware.

Research Intern: Deep Learning & Computer Vision

K.J. Somaiya School of Engineering

May 2025 – Jul 2025

Mumbai, Maharashtra

- Developed an AI-powered computer vision platform for automated archaeological rock art analysis using Python, TensorFlow, OpenCV, Streamlit, FAISS, and SQLite.
- Designed an end-to-end pipeline featuring U-Net-based image segmentation, deep feature extraction, similarity search, batch annotation, and dataset management, achieving 89.2% IoU and reducing manual annotation time by 40%.

RESEARCH PUBLICATIONS

Automated Segmentation for Archaeological Rock Art: A Comparative Pipeline Study

IDEA 2026

Springer Nature Lecture Notes in Networks and Systems (LNNS) | Paper ID: 297 | Top 15% of submissions

Authors: Sunayana V. Jadhav, Irfan Siddavattam, Ashwini Dalvi, Aayush Solanke, Aryan Lodha

- Developed a comprehensive computer vision pipeline comparing six segmentation approaches (Canny, Watershed, Mask R-CNN, U-Net, SAM) on 1,976 high-resolution rock art images from Bhimbetka and Saru-Maru Cave heritage sites.
- Achieved 89.2% IoU with domain-specific U-Net + post-processing and 40% reduction in manual annotation time; led feature extraction, model implementation, comparative analysis, and results interpretation.

PROJECTS

LLM DevOps Orchestrator Suite | Python, FastAPI, LangGraph, Multi-Agent AI, LLM APIs

- Architected a Multi-Agent LLM orchestration platform routing software engineering tasks to the most suitable model by complexity, cost, and performance, with 14 specialized agents for prompt optimization, workflow orchestration, evaluation, monitoring, and analytics.

Text-to-SQL Interface with Guardrails and Hallucination Detection | Python, FastAPI, PostgreSQL, SQLAlchemy, LLM APIs, Docker

- Engineered a production-ready Text-to-SQL platform with schema-aware prompting, configurable safety guardrails, LLM-based hallucination detection, multi-query verification, and confidence scoring for secure, explainable database querying.

AutoMind AI Deployment Engine | Python, FastAPI, Docker, MLOps, Machine Learning

- Built an end-to-end MLOps platform automating data preprocessing, model training, hyperparameter optimization, API generation, containerized deployment, and production monitoring via scalable FastAPI/Docker pipelines.

Autonomous Cyber Defense AI Agent | Python, Multi-Agent AI, Machine Learning, Cybersecurity

- Engineered a Multi-Agent AI cybersecurity platform with agents for anomaly detection, behavioral analysis, vulnerability assessment, and automated network isolation for real-time threat detection and response.

MarketMind AI | React, Node.js, MongoDB, Ollama LLM, Puppeteer, Docker

- Engineered an autonomous multi-agent market intelligence platform translating unstructured e-commerce data from Amazon, Flipkart, and Reddit into competitive blueprints via local LLM synthesis, fault-tolerant scraping, and SSE-based real-time streaming.

SKILLS

Languages & Databases: Python, JavaScript, SQL, PostgreSQL, MongoDB, SQLite, SQLAlchemy

Machine Learning & Computer Vision: PyTorch, TensorFlow, Scikit-learn, XGBoost, OpenCV, YOLOv8, ByteTrack, U-Net

AI/LLM & Web Development: Multi-Agent AI Systems, LLM APIs, LangGraph, MLOps, FAISS, FastAPI, React, Node.js, MERN Stack, REST APIs, Docker, Streamlit

PROFESSIONAL CERTIFICATIONS

Google Cloud Data Analytics - Google • The Web Developer Bootcamp 2026 - Udemy (Colt Steele) • Data Analysis with Python - freeCodeCamp • Problem Solving (Intermediate) - HackerRank • Software Engineer - HackerRank • SQL (Advanced) - HackerRank • Career Essentials in Generative AI - Microsoft & LinkedIn Learning